CITS3402 - High Performance Computing
Assignment 1: Report

The University of Western Australia

James Oakey (22709404) Brent Tankink (23842541)

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1 Introduction

2 Birthday-attack algorithm

2.1 Search strategy

2.2 Expected number of trials

2.3 Collision-detection data structure

3 Parallelisation with OpenMP

3.1 Decomposition

3.2 Race conditions on the shared table

3.3 Other correctness concerns

4 Memory usage and trade-offs

5 Performance evaluation

5.1 Methodology

5.2 Speedup and scalability

5.3 Discussion

6 Conclusion

A Reproducing the results